

WEEK 01- STATUS

Planned Work

Formed a team. Set up a github repo, and added a every team members.

Do some research or study about STC89C52 microcontroller

Designed a initial system architecture and selected a programming language.

Completed Work

The team was formed and complete github set up

Programming language was finalized

Initial system architecture was done and discussion about the project was also done

Completed of creating files and folders like docx(decisions,meetings,weekly status), gitignore, README, src, program, tests.

Meeting -1 (22-8-2026)

Participants: VENKATESHWARA U D

RAAIF ABDUL HAAMID

ZAINABA FIDHA K N

YATHIKSHA U S

Agenda:

1.About Problem Statement.

2.Work Distribution and Git hub repo.

Meeting -2 (28-8-2026)

Participants: VENKATESHWARA U D

RAAIF ABDUL HAAMID

ZAINABA FIDHA K N

YATHIKSHA U S

Agenda:

- 1) Finalize individual research topics (STC89C52 architecture, memory & stack, data structures, OS scheduling & context switching).**
- 2) Review GitHub repo structure and confirm folders/files (docx, gitignore, README, src, program, tests)**

Decision Taken

Venkateshwara studied and researched about STC89C52 8-bit microcontroller.

Fida studied and researched about memory and stack, PC

Yathiksha researched and studied about queue, stack and DSA

Raaif researched and studied about the context switching and OS, GPIO

Actions taken:

Venkateshwara U D -25190152(STC89C52 & CPU Architecture): Studied and documented the STC89C52's CPU architecture -ALU, Control Unit, registers (ACC, B, PSW, R0–R7), Program Counter, Stack Pointer, and DPTR. Documented the Fetch–Decode–Execute–Write Back instruction cycle with a worked example, the interrupt system (INT0, INT1, Timer 0/1, Serial with IE/IP registers), and GPIO input/output behavior across Ports 0–3.

Zainaba Fidha K N-25190155 (Memory & Stack): Documented the STC89C52's memory organization program memory vs. data memory, the 128-byte internal RAM layout (register banks, bit-addressable area, general-purpose RAM), and the SFR region. Worked through the stack mechanism in detail: how the Stack Pointer moves on PUSH (increment-then-store) vs. POP (read-then-decrement), and covered CALL/RET behavior along with the stack overflow risk.

Yathiksha U S -25190153 (Data Structures & Process Management): Documented the core data structures for the simulator -arrays, queues, circular queues, and the PCB (Process ID, state, PC, priority). Mapped out the process lifecycle (New → Ready → Running → Waiting/Terminated) and the ready queue, plus a 3-level testing strategy (component, module-integration, full-simulator).

Raaif Abdul Haamid -25190140(OS Scheduling & Context Switching): Studied and documented timers and interrupts and how they trigger scheduling, then covered FCFS, Round Robin, and Priority scheduling with worked examples. Documented context switching

(saving/restoring PC, registers, flags) and waiting/turnaround/response time as metrics for comparing algorithms, and mapped how all four members' modules connect.

Work Carried Forward to Week2

Complete the remaining research and documentation on the STC89C52 microcontroller, including CPU architecture, registers, instruction execution, memory, GPIO, timers, and interrupts. Finalize the integration of the CPU, data structures, and process scheduling components with the team.

Week 2 Plan

- Consolidate all Week 1 research into a single architecture reference doc.
- Assign roles and what and how we should code and the requirements to continue in this project.
- Understand the working of the source code and how we should simulate it.